



ABBOTSLEIGH

AUGUST 2007

**YEAR 12
ASSESSMENT 4
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks **(120)**

- Attempt Questions 1-10.
- All questions are of equal value.

Outcomes assessed

Preliminary course

- P1** demonstrates confidence in using mathematics to obtain realistic solutions to problems.
- P2** provides reasoning to support conclusions that are appropriate to the context
- P3** performs routine arithmetic and algebraic manipulation involving surds, simple rational expressions and trigonometric identities
- P4** chooses and applies appropriate arithmetic, algebraic, graphical, trigonometric and geometric techniques
- P5** understands the concept of a function and the relationship between a function and its graph
- P6** relates the derivative of a function to the slope of its graph
- P7** determines the derivative of a function through routine application of the rules of differentiation
- P8** understands and uses the language and notation of calculus

HSC course

- H1** seeks to apply mathematical techniques to problems in a wide range of practical contexts
- H2** constructs arguments to prove and justify results
- H3** manipulates algebraic expressions involving logarithmic and exponential functions
- H4** expresses practical problems in mathematical terms based on simple given models
- H5** applies appropriate techniques from the study of calculus, geometry, probability, trigonometry and series to solve problems
- H6** uses the derivative to determine the features of the graph of a function
- H7** uses the features of a graph to deduce information about the derivative
- H8** uses techniques of integration to calculate areas and volumes
- H9** communicates using mathematical language, notation, diagrams and graphs

Question 1 (12 marks)**Use a SEPARATE writing booklet**

(a) Evaluate $\sqrt{\frac{85.7}{6.8 \times 5.2}}$ correct to 2 significant figures. 2

(b) Evaluate $\sum_{r=3}^5 r^2$. 1

(c) Solve $|3x-1| \geq 5$. 2

(d) Rationalise the denominator of $\frac{2}{2\sqrt{3}+5}$. 2

(e) Simplify $\frac{2}{b-3} \div \frac{6}{b^2-9}$. 2

(f) Solve $\frac{2x-1}{3} - \frac{x-2}{2} = 5$. 3

Question 2 (12 marks)
Use a SEPARATE writing booklet

(a) Find the coordinates of the vertex of the parabola $(y-4)^2 = 8(x+3)$. 1

(b) Differentiate with respect to x :

(i) $\frac{x^2}{x-1}$ 2

(ii) $x \cos x$ 2

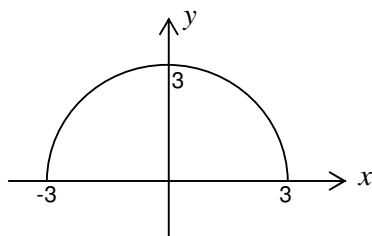
(iii) $\log_e(x^3-1)$ 2

(c) Find $\int \frac{x^3-4}{x} dx$ 2

(d) Evaluate $\int_0^3 6e^{2x} dx$ leaving your answer in exact form. 3

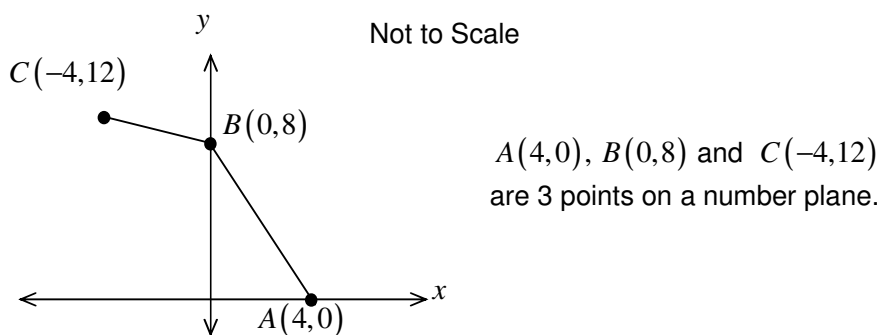
Question 3 (12 marks)
Start a new booklet

(a)



- | | | |
|-------|---|----------|
| (i) | State the equation of the semicircle shown above. | 1 |
| (ii) | State the domain of this function. | 1 |
| (iii) | State the range of this function. | 1 |

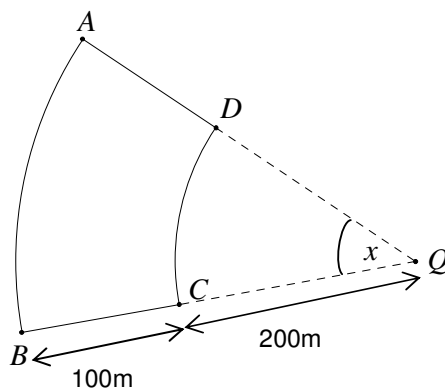
(b)



- | | | |
|-------|---|----------|
| (i) | Draw the diagram into your answer booklet and find the length of AB in simplest surd form. | 2 |
| (ii) | Find the gradient of the line AB . | 1 |
| (iii) | The line l is parallel to AB and passes through C . Show that the equation of l is $2x + y - 4 = 0$. | 2 |
| (iv) | Draw l on your diagram and find the coordinates of D , the point where l intersects the y axis. | 1 |
| (v) | Find the perpendicular distance from B to the line l . | 2 |
| (vi) | Find the area of parallelogram $ABCD$. | 1 |

Question 4 (12 marks)
Start a new booklet

(a)



In the figure AB and CD are circular arcs which subtend an angle x radians at the centre Q where $0 < x < \pi$ and AQ, BQ are radii. The length BC is 100m and CQ is 200m.

- (i) Find expressions in terms of x for the length of the arcs AB and CD . 1
- (ii) A man lives at A and there is a bus stop at B , with the paths AB, BC, CD and DA in the figure forming a road system. For what values of x is it shorter for the man to walk along route $ADCB$ rather than along the arc AB ? 2
- (iii) Given $x = \frac{\pi}{5}$ find the area enclosed by the roads linking A, B, C and D . 2
- (b) (i) Write down the discriminant of $x^2 - 2kx + 6k$. 2
- (ii) For what values of k is $x^2 - 2kx + 6k$ positive definite? 2
- (c) Find the volume of the solid of revolution formed by rotating the curve $y = x + \frac{1}{x}$ about the x axis between $x = 1$ and $x = 3$. 3

Question 5 (12 marks)

Start a new booklet

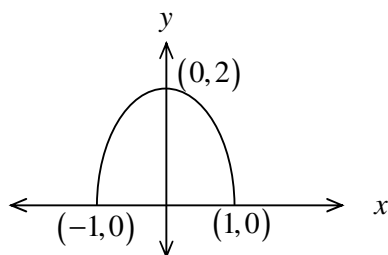
- (a) The fifth term of an arithmetic series is 14 and the sum of the first 10 terms is 165. Find the first term of the series. **3**
- (b) Use Simpson's rule with 3 function values to estimate $\int_0^1 4^x dx$. **2**
- (c) From the integers 1 to 25, one integer is chosen at random. What is the probability that it is:
- (i) divisible by 5 and greater than 18. **1**
- (ii) divisible by 5 or greater than 18. **1**
- (d) The roots of the quadratic equation $px^2 - x + q = 0$ are -1 and 3 . Find p and q . **2**
- (e) The curve $y = ax^3 - 9x^2 + b$ has a minimum turning point at $(3, -12)$. Find the values of a and b . **3**

Question 6 (12 marks)
Start a new booklet

(a) Solve $\cos \theta = \frac{-\sqrt{3}}{2}$ for $0 \leq \theta \leq 2\pi$

2

(b)



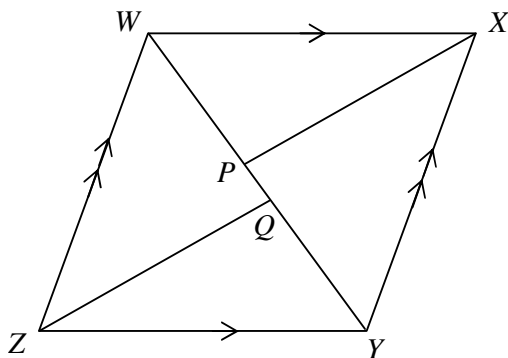
An ornamental arch window 2m wide and 2m high is to be made in the shape pictured on the axes pictured.

The architect decided that $y = 2 \cos \frac{\pi}{2} x$ or $y = 2 - 2x^2$ would be suitable equations to use.

By finding the area of each possible window, determine which one would allow the greatest amount of light in.

4

(c)



$WXYZ$ is a parallelogram.
 XP bisects $\angle WXY$ and
 ZQ bisects $\angle WZY$.
 WY is a diagonal.

(i) Copy the diagram into your answer booklet and explain why $\angle WXY = \angle WZY$.

1

(ii) Prove $\triangle WXP \equiv \triangle YZQ$.

3

(iii) Hence find the length of PQ given $WY = 20\text{cm}$ and $QY = 8\text{cm}$.

2

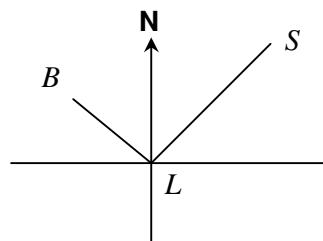
Question 7 (12 marks)

Start a new booklet

- (a) (i) Sketch on the same number plane the graphs of $y = 3\sin 2x$ and $y = 1 - \cos x$ for $0 \leq x \leq 2\pi$. 3

- (ii) Using your graphs, determine the number of solutions the equation $3\sin 2x + \cos x = 1$ will have in the given domain. 1

- (b) From a lighthouse L a ship S bears 053° and is at a distance of 8 nautical miles. From L a boat B bears 293° and is at a distance of 6 nautical miles.

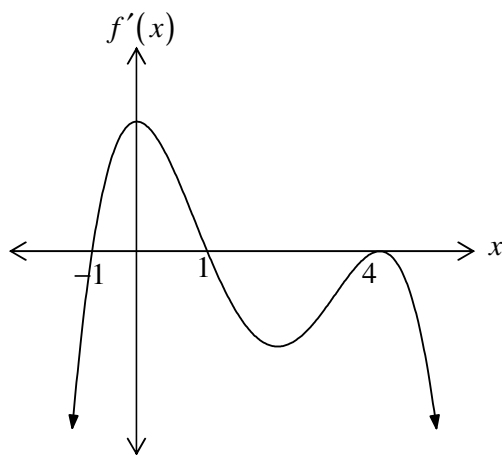


- (i) Draw the diagram in your answer booklet and mark on the given information. 1

- (ii) Find the distance of ship S from Boat B . Give your answer as a surd. 2

- (iii) Find the bearing of ship S from Boat B . Give your answer to the nearest degree. 2

(c)



Not to Scale

The graph of $y = f'(x)$ is sketched above.

- (i) Write down the values of x where stationary points will occur in the graph of $y = f(x)$. 1

- (ii) Sketch $y = f(x)$ given that it passes through $(0,0)$ and $(4,-2)$. Clearly show any turning points or points of inflection. 2

Question 8 (12 marks)
Start a new booklet

- (a) If $y = e^{4x} + e^{2x}$
- (i) Find $\frac{dy}{dx}$ 1
- (ii) Show that $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 8y = 0$ 2
-
- (b) The mass M in grams of a radioactive substance may be expressed as $M = Ae^{-kt}$ where t is the time in years and k is a constant.
- (i) If initially the mass is 10 grams, find A . 1
- (ii) After 5 years the mass is 9 grams. Find the mass after 20 years. 2
-
- (c) A gold coin is biased so that the probability of tossing a head is $\frac{2}{3}$. A silver coin is 'fair' with an equal chance of tossing a head or a tail.
- (i) The two coins are tossed together. what is the probability of tossing
- (α) 2 heads. 1
- (β) a head and a tail. 2
- (γ) at least 1 tail. 1
- (ii) One of the coins is selected at random and then tossed twice. What is the probability of tossing 2 tails. 2

Question 9 (12 marks)
Start a new booklet

- (a) The line $y = mx$ is a tangent to the curve $y = e^{3x}$. Find the value of m . **2**
- (b) Mary decided to contribute to a superannuation fund by investing \$3000 each year at the beginning of 1990. The interest was paid at a rate of 7% p.a. for 10 years after which it was changed to 8% p.a. starting January 1, 2000. She plans to continue to invest in this fund until the year 2010.
- (i) Show that after 10 years her investment is worth \$44 350.80. **2**
- (ii) How much is her investment worth at the end of the 11th year? **1**
- (iii) How much is her investment worth at the end of 2010? **2**
- (c) The cost of running a long distance transport truck is $\$ \left(\frac{V^2}{3} + 600 \right)$ PER HOUR where V is the speed in kilometres per hour.
- (i) Find an expression for the number of hours travelled over a distance of 100 km. **1**
- (ii) Show that the Total Cost of running the truck for 100km is given by
- $$C = \frac{100V}{3} + \frac{60000}{V}$$
- 1**
- (iii) Using the expression in part (ii) find the most economical speed for running the truck for 100 km at that speed. (i.e. the speed which gives the minimum cost) **3**

Question 10 (12 marks)
Start a new booklet

- (a) (i) For what values of x does the geometric series $1 + 4x^2 + 16x^4 + 64x^6 + \dots$ have a limiting sum? **2**

- (ii) Find the limiting sum when $x = \frac{1}{3}$. **2**

- (b) A rainwater tank with a volume of 9m^3 is installed in a new house. At 8am rain begins to fall and flows into the empty tank at the rate given by

$$\frac{dV}{dt} = \frac{36t}{t^2 + 20}$$

where t is the time in hours and V is the volume measured in cubic metres ($t = 0$ is represented by 8am).

- (i) Show that the volume of water in the tank at time t is given by

$$V = 18 \log_e \left(\frac{t^2 + 20}{20} \right), \quad t > 0. \quad \text{2}$$

- (ii) Find the time when the tank will be completely filled with water (to the nearest minute). **3**

- (iii) Later, when the tank is full and the rain has stopped, Louise turns on the pump which pumps the water out at the rate given by $\frac{dV}{dt} = \frac{t^2}{k}$. The pump continues for 5 hours until the tank is empty. Find the value of k . **3**

END OF PAPER

$$\textcircled{1} \text{ (a) } \sqrt{\frac{85.7}{6.8 \times 5.2}} = 1.55680$$

$$= \underline{\underline{1.6}} \text{ to 2 sign figs}$$

$$\text{(b) } \sum_{r=3}^5 r^2 = 3^2 + 4^2 + 5^2$$

$$= \underline{\underline{50}}$$

$$\text{(c) } 3x-1 \leq -5 \quad \text{or} \quad 3x-1 \geq 5$$

$$3x \leq -4 \qquad 3x \geq 6$$

$$\underline{\underline{x \leq -\frac{4}{3} \quad \text{or} \quad x \geq 2}}}$$

$$\text{(d) } \frac{2}{2\sqrt{3}+5} \times \frac{2\sqrt{3}-5}{2\sqrt{3}-5}$$

$$= \frac{4\sqrt{3}-10}{12-25} = \frac{4\sqrt{3}-10}{-13} = \underline{\underline{\frac{10-4\sqrt{3}}{13}}}$$

$$\text{(e) } \frac{2}{b-3} \div \frac{6}{b^2-9} = \frac{2}{b-3} \times \frac{b^2-9}{6}$$

$$= \frac{2}{\cancel{b-3}} \times \frac{(b+3)\cancel{(b-3)}}{6}$$

$$= \underline{\underline{\frac{b+3}{3}}}$$

$$\text{(f) } \frac{2x-1}{3} - \frac{x-2}{2} = 5$$

$$2(2x-1) - 3(x-2) = 30$$

$$4x-2-3x+6=30$$

$$x+4=30$$

$$\underline{\underline{x=26}}$$

$$\textcircled{2} \text{(a) } (y-4)^2 = 8(x+3) \text{ has vertex } \underline{\underline{(-3, 4)}}$$

$$\text{(b) (i) } \frac{d}{dx} \left(\frac{x^2}{x-1} \right) = \frac{(x-1)2x - x^2}{(x-1)^2}$$

$$= \frac{2x^2 - 2x - x^2}{(x-1)^2}$$

$$= \underline{\underline{\frac{x^2 - 2x}{(x-1)^2}}}$$

$$\text{(ii) } \frac{d}{dx} (x \cos x) = x \times -\sin x + \cos x$$

$$= \underline{\underline{\cos x - x \sin x}}$$

$$\text{(iii) } \frac{d}{dx} (\log_e (x^3-1)) = \underline{\underline{\frac{3x^2}{x^3-1}}}$$

$$\text{(c) } \int \frac{x^3-4}{x} dx = \int \frac{x^3}{x} - \frac{4}{x} dx$$

$$= \int x^2 - \frac{4}{x} dx$$

$$= \underline{\underline{\frac{x^3}{3} - 4 \log_e x}}$$

$$\text{(d) } \int_0^3 6e^{2x} dx = \left[\frac{6e^{2x}}{2} \right]_0^3$$

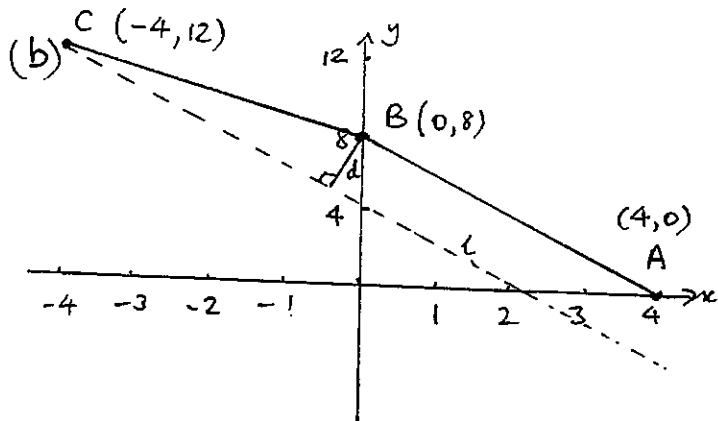
$$= 3(e^6 - e^0)$$

$$= \underline{\underline{3(e^6 - 1)}}$$

③ (a) (i) $y = \sqrt{9-x^2}$ is eqn of semicircle

(ii) Domain : $-3 \leq x \leq 3$

(iii) Range : $0 \leq y \leq 3$



(i) $AB = \sqrt{4^2 + 8^2}$ (By Pythagoras or distance formula)
 $= \sqrt{80}$
 $= 4\sqrt{5}$

(ii) $m = \frac{0-8}{4-0} = -2$

(iii) $y - y_1 = m(x - x_1)$
 $y - 12 = -2(x + 4)$
 $y - 12 = -2x - 8$
 $2x + y - 4 = 0$

(iv) When $x=0$ $0 + y - 4 = 0$
 $y = 4$

$D(0,4)$

(v) $d = \frac{|2x + y - 4|}{\sqrt{2^2 + 1^2}}$
 $= \frac{|2 \times 0 + 8 - 4|}{\sqrt{5}}$
 $= \frac{4}{\sqrt{5}}$ units

(vi) $A = bh$
 $= AB \times d$
 $= 4\sqrt{5} \times \frac{4}{\sqrt{5}}$
 $= 16$ square units

④ (a) (i) $l = r\theta$

$AB = 300x$

$CD = 200x$

(ii) Route ADCB = $100 + 200x + 100$
 $= 200 + 200x$

$200 + 200x < 300x$

$200 < 100x \therefore x > 2$

$2 < x < \pi$ radians (since $0 < x < \pi$ by defn).

(iii) Area ADCB = Area sector AQB - Area sector DQB
 $= \frac{1}{2} \times 300^2 x \left(\frac{\pi}{5}\right) - \frac{1}{2} \times 200^2 x \frac{\pi}{5}$
 $= \frac{\pi}{10} (90000 - 40000)$
 $= 5000\pi$ square units

(b) (i) $x^2 - 2kx + 6k$
 $\Delta = b^2 - 4ac$
 $= (-2k)^2 - 4 \times 1 \times 6k$
 $= 4k^2 - 24k$

(ii) positive definite means $\Delta < 0$
 $4k^2 - 24k < 0$
 $4k(k - 6) < 0$
 $0 < k < 6$

(c) $V = \pi \int_1^3 y^2 dx$
 $= \pi \int_1^3 \left(x + \frac{1}{x}\right)^2 dx$
 $= \pi \int_1^3 x^2 + 2 + \frac{1}{x^2} dx$
 $= \pi \int_1^3 x^2 + 2 + x^{-2} dx$
 $= \pi \left[\frac{x^3}{3} + 2x - x^{-1} \right]_1^3$
 $= \pi \left[\left(\frac{27}{3} + 6 - \frac{1}{3}\right) - \left(\frac{1}{3} + 2 - 1\right) \right]$
 $= \pi \left(\frac{44}{3} - \frac{4}{3} \right)$
 $= \frac{40\pi}{3}$ cubic units

$$(5) (a) T_5 = 14$$

$$S_{10} = 165$$

$$a + 4d = 14 \quad (1)$$

$$\frac{10}{2} (2a + 9d) = 165 \quad (2)$$

$$5(2a + 9d) = 165$$

$$2a + 9d = 33 \quad (3)$$

$$(1) \times 2: 2a + 8d = 28 \quad (4)$$

$$(3) - (4) \quad d = 5$$

$$\text{sub in (1)} \quad a + 4 \times 5 = 14$$

$$\underline{\underline{a = -6}}$$

$$(b) \begin{array}{c|c|c|c} x & 0 & \frac{1}{2} & 1 \\ \hline 4^x & 1 & 2 & 4 \end{array}$$

$$\int_0^1 4^x dx = \frac{1-0}{6} (1 + 4 \times 2 + 4)$$

$$= \frac{1}{6} \times 13$$

$$= \underline{\underline{\frac{13}{6}}}$$

$$(c) (i) \div \text{by } 5 \text{ and } > 18 \text{ is } 20, 25$$

$$\therefore \text{Prob} = \underline{\underline{\frac{2}{25}}}$$

$$(ii) \div \text{by } 5 \text{ or } > 18 \quad 5, 10, 15, 20, 25, 19, 21, 22, 23, 24$$

$$\therefore \text{Prob} = \frac{10}{25} = \underline{\underline{\frac{2}{5}}}$$

$$(d) \quad px^2 - x + q = 0 \quad \text{roots } -1, 3$$

$$\text{sum of roots} = -\frac{b}{a} = \frac{1}{p}$$

$$-1 + 3 = \frac{1}{p} \quad \therefore 2 = \frac{1}{p} \quad \underline{\underline{p = \frac{1}{2}}}$$

$$\text{product of roots} \quad -1 \times 3 = \frac{q}{p}$$

$$\frac{1}{2} \times -3 = q \quad \underline{\underline{q = -\frac{3}{2}}}$$

$$(e) \quad y = ax^3 - 9x^2 + b$$

$$y' = 3ax^2 - 18x$$

$$\text{When } x = 3 \quad y' = 0 \quad 0 = 27a - 54$$

$$27a = 54$$

$$\underline{\underline{a = 2}}$$

$$y = 2x^3 - 9x^2 + b$$

$$\text{sub in } x = 3 \quad y = -12 \quad (\text{lies on curve})$$

$$-12 = 2 \times 27 - 9 \times 9 + b$$

$$-12 = 54 - 81 + b$$

$$-12 = -27 + b$$

$$\underline{\underline{b = 15}}$$

$$(6) (a) \cos \theta = -\frac{\sqrt{3}}{2}$$

$$0 \leq \theta \leq 2\pi$$

$$\text{Basic angle} = \frac{\pi}{6}$$

$$\theta = \pi - \frac{\pi}{6}, \pi + \frac{\pi}{6}$$

$$\begin{array}{c} \frac{\pi}{6} \\ \swarrow \quad \searrow \\ \frac{\pi}{2} \end{array} \quad \begin{array}{l} \text{2nd} \\ \text{3rd} \end{array} \text{quad}$$

$$\underline{\underline{\theta = \frac{5\pi}{6}, \frac{7\pi}{6}}}$$

$$(b) \quad 2 \int_0^1 2 \cos \frac{\pi}{2} x dx$$

$$= 4 \int_0^1 \cos \frac{\pi}{2} x dx = 4 \left[\frac{2 \sin \frac{\pi}{2} x}{\pi} \right]_0^1$$

$$= \frac{8}{\pi} (\sin \frac{\pi}{2} - \sin 0)$$

$$= \frac{8}{\pi} \div 2.5464 \text{ square units}$$

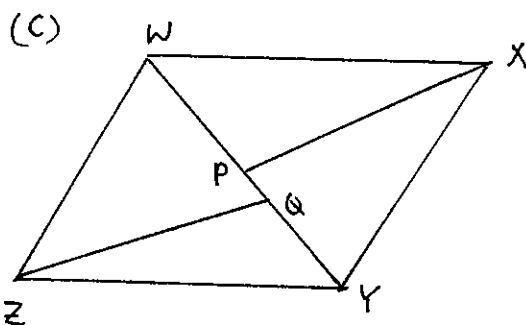
$$2 \int_0^1 2 - 2x^2 dx = 4 \int_0^1 1 - x^2 dx$$

$$= 4 \left[x - \frac{x^3}{3} \right]_0^1$$

$$= 4 \left[\left(1 - \frac{1}{3}\right) - (0 - 0) \right]$$

$$= \frac{8}{3} = \underline{\underline{2\frac{2}{3}}} = 2.6$$

The parabolic equation would let more light in.



(i) $\angle WXY = \angle WZY$ (opposite angles of a parallelogram are equal)

(ii) In $\triangle WXP, \triangle YZQ$

$\angle WXP = \frac{1}{2} \angle WXY$ (XP bisects $\angle WXY$)

$\angle QZY = \frac{1}{2} \angle WZY$ (ZQ bisects $\angle WZY$)

since $\angle WXY = \angle WZY$ (from (i)) $\angle WXP = \angle QZY$

$WX = ZY$ (opposite sides of parallelogram)

$\angle XWP = \angle ZYQ$ (alternate angles, $WX \parallel ZY$)

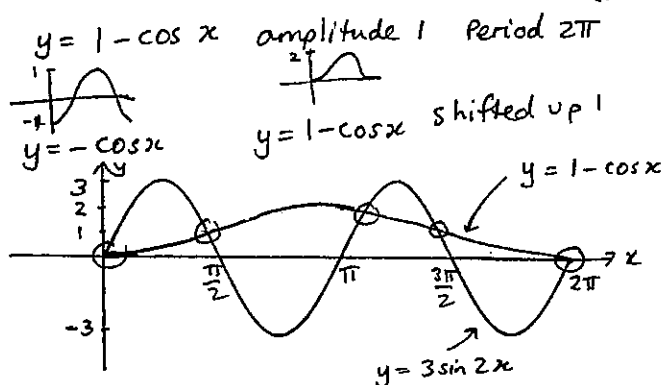
$\therefore \triangle WXP \equiv \triangle YZQ$ (AAS)

(ii) Since $QY = 8$, $WP = 8$ (corresponding sides in congruent triangles)

so $PQ = 20 - (8 + 8)$

$$\underline{\underline{PQ = 4 \text{ cm}}}$$

⑦ (a) (i) $y = 3 \sin 2x$ amplitude 3 period π

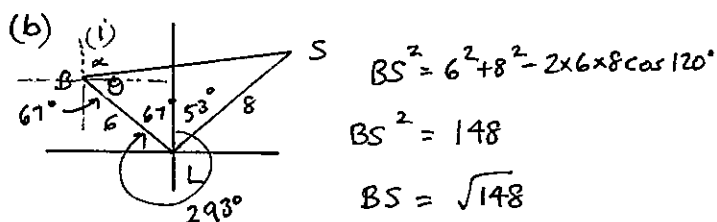


(ii) $3 \sin 2x + \cos x = 1$

$3 \sin 2x = 1 - \cos x$

Solution will be the number of points of intersection between the graphs $y = 3 \sin 2x$ and $y = 1 - \cos x$

There are 5 solutions for $0 \leq x \leq 2\pi$



(iii) Need to find θ

$\frac{\sin \theta}{8} = \frac{\sin 120^\circ}{\sqrt{148}}$

$\sin \theta = \frac{8 \sin 120^\circ}{\sqrt{148}}$

$= 0.56949$

$\theta = 34^\circ 43' = 35^\circ$

$\alpha = 180 - 35^\circ - 67^\circ$
 $= 78^\circ$

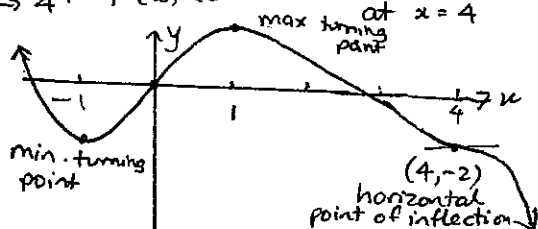
$\therefore$ Bearing is 078°

(c) (i) stat pts occur when $f'(x) = 0$
 ie. when $x = -1, 1, 4$

(ii) As $x \rightarrow -1^-$ $f'(x) < 0$ $\searrow$
 $x \rightarrow -1^+$ $f'(x) > 0 \therefore$ minimum at $x = -1$

As $x \rightarrow 1$ $f'(x) > 0$
 $x \rightarrow 1^+$ $f'(x) < 0 \therefore$ maximum at $x = 1$

As $x \rightarrow 4^-$ $f'(x) < 0$ $\searrow$ $\therefore$ point of inflection
 $x \rightarrow 4^+$ $f'(x) < 0$ $\searrow$ $\therefore$ point of inflection



⑧ (a) (i) $y = e^{4x} + e^{2x}$
 $\frac{dy}{dx} = 4e^{4x} + 2e^{2x}$

(ii) $\frac{d^2y}{dx^2} = 16e^{4x} + 4e^{2x}$

LHS $= \frac{d^2y}{dx^2} - 6 \frac{dy}{dx} + 8y$
 $= 16e^{4x} + 4e^{2x} - 6(4e^{4x} + 2e^{2x}) + 8(e^{4x} + e^{2x})$
 $= 16e^{4x} + 4e^{2x} - 24e^{4x} - 12e^{2x} + 8e^{4x} + 8e^{2x}$
 $= 0$
 $= \text{RHS.}$

(b) (i) $M = Ae^{-kt}$

$t = 0$ $M = 10$ $10 = Ae^0$
 $\therefore A = 10$

(ii) $t = 5$ $M = 9$

$9 = 10e^{-k \times 5}$

$\frac{9}{10} = e^{-5k}$

$\ln \frac{9}{10} = -5k$

$k = -\frac{1}{5} \ln \left(\frac{9}{10} \right) \div 0.0210721$
 -20×0.0210721

When $t = 20$ $M = 10e$
 $= 6.561$

Mass is 6.561 grams

(c) (i) (a) $P(HH) = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$

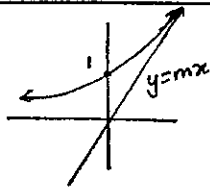
(b) $P(HT) + P(TH) = \frac{2}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{2}$
 $= \frac{1}{3} + \frac{1}{6}$
 $= \frac{1}{2}$

(c) $P(\text{at least 1 tail}) = 1 - P(HH)$
 $= 1 - \frac{1}{3}$
 $= \frac{2}{3}$

(ii) $P(\text{gold coin, T, T}) + P(\text{silver coin, T, T})$

$= \frac{1}{2} \times \frac{1}{3} \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$
 $= \frac{1}{18} + \frac{1}{8}$
 $= \frac{13}{72}$

(9) (a)



$$y = e^{3x}$$

$$\frac{dy}{dx} = 3e^{3x}$$

gradient of $y = mx$ is m

so $m = 3e^{3x}$

line & curve intersect when $mx = e^{3x}$

But $m = 3e^{3x}$ so $3e^{3x}x = e^{3x}$

($\div$ by e^{3x}) $3x = 1$
 $x = \frac{1}{3}$

$\therefore$ intersect at $x = \frac{1}{3}$

when $x = \frac{1}{3}$ $m = 3e^{3 \times \frac{1}{3}}$
 $m = 3e$

(b) end 1990 1st 3000 amounts to $3000(1.07)^1$
 1991 2nd 3000 " " $3000(1.07)^2$
 1999 10th 3000 " " $3000(1.07)^{10}$

geometric series
 $3000(1.07)^1 + 3000(1.07)^2 + \dots + (1.07)^{10}$
 $\text{Total} = \frac{a(r^n - 1)}{r - 1} = \frac{3000(1.07)(1.07^{10} - 1)}{1.07 - 1}$

$= \$44\,350.80$

(ii) At end of 11th year

Amt = $44\,350.80(1.08) + 3000(1.08)$
 $= \$51\,138.86$

(or $47\,350.8 \times 1.08$)

(iii) $\$44\,350.80$ earns 8% compound interest for 11 years (2000 - 2010)

$\$3000$ invested every year at 8% for 11 years
 $\text{Total} = 44\,350.80(1.08)^{11} + \frac{3000(1.08)(1.08^{11} - 1)}{1.08 - 1}$

$= 103\,410.05 + 53\,931.38$

$= \$157\,341.43$

(c) (i) Speed = $\frac{\text{Dist}}{\text{Time}}$ so Time = $\frac{D}{S}$
 $= \frac{100}{V}$

(ii) Total cost = cost/hr $\times$ no. of hours
 $= \left(\frac{V^2}{3} + 600\right) \times \frac{100}{V}$
 $C = \frac{100V}{3} + \frac{60000}{V}$

(iii) $\frac{dC}{dV} = \frac{100}{3} - 60000V^{-2}$

Let $\frac{dC}{dV} = 0$ $\frac{100}{3} - \frac{60000}{V^2} = 0$

$\frac{100}{3} = \frac{60000}{V^2}$

$V^2 = \frac{180000}{100} = 1800$

$V = \sqrt{1800} = 30\sqrt{2}$ km/hr
 (take positive)

$\frac{d^2C}{dV^2} = +\frac{120000}{V^3}$

when $V = 30\sqrt{2}$ $\frac{d^2C}{dV^2} = \frac{120000}{(30\sqrt{2})^3} > 0$ $\therefore V$

$\therefore V = 30\sqrt{2}$ gives the minimum cost

$V \doteq 42.4$ km/hr

(10) (a) (i) Limiting sum when $-1 < r < 1$

$1 + 4x^2 + 16x^4 + 64x^6 + \dots$

$r = 4x^2$

so $-1 < 4x^2 < 1$

$4x^2$ is always positive so just need to solve $4x^2 < 1$

$x^2 < \frac{1}{4}$

$x^2 - \frac{1}{4} < 0$

$(x + \frac{1}{2})(x - \frac{1}{2}) < 0$

$-\frac{1}{2} < x < \frac{1}{2}$



(ii) when $x = \frac{1}{3}$ $S_{\infty} = \frac{a}{1-r}$
 $4x^2 = \frac{4}{9}$ $= \frac{1}{1 - \frac{4}{9}}$
 $= \frac{9}{5}$

(b) (i) $\frac{dV}{dt} = \frac{36t}{t^2 + 20}$

$V = \int \frac{36t}{t^2 + 20} dt$

$= 18 \int \frac{2t}{t^2 + 20} dt$

$V = 18 \log_e(t^2 + 20) + C$

$t = 0$ $V = 0$ $0 = 18 \log_e(0 + 20) + C$

$C = -18 \log_e(20)$

$V = 18 \log_e(t^2 + 20) - 18 \log_e 20$

$= 18 \log_e \left(\frac{t^2 + 20}{20} \right)$

(ii) Tank will be filled when $V = 9$

$9 = 18 \log_e \left(\frac{t^2 + 20}{20} \right)$

$\frac{1}{2} = \log_e \left(\frac{t^2 + 20}{20} \right)$

$\frac{t^2 + 20}{20} = e^{\frac{1}{2}}$

$t^2 = 20e^{\frac{1}{2}} - 20$

$t^2 = 12.9744$

$t = \sqrt{12.9744}$
 take +ve for time

$t = 3.6$ hours

$= 3$ hrs 36 mins

(iii) $V = \int \frac{t^2}{k} dt$

$V = \frac{t^3}{3k} + C$

$t = 0$ $V = 9$ $9 = 0 + C$

$V = \frac{t^3}{3k} + 9$

But when $t = 5$ $V = 0$

$0 = \frac{5^3}{3k} + 9$

$-9 = \frac{125}{3k}$

$k = -\frac{125}{27}$

or
 $V = \int_0^5 \frac{t^2}{k} dt$

$9 = \left[\frac{t^3}{3k} \right]_0^5$

$9 = \left(\frac{125}{3k} - 0 \right)$

$3k = \frac{125}{9}$

$k = \frac{125}{9}$

emptying so ma
 $= -\frac{125}{9}$